STATISTICS & MACHINE LEARNING · LESSON 6 OF 30

A/B Testing and Experimentation: Design, Power, Analysis and Pitfalls

Run product experiments the way data teams at e-commerce and fintech companies do. Choose a metric and guardrails, size the test with power analysis, randomise, analyse conversion and revenue with z-tests, t-tests and bootstrap intervals, check sample-ratio mismatch, avoid peeking and novelty effects, and write the decision memo, using a new-checkout experiment.

ML class 2 (case study) · ✦ statsmodels 0.1x · SciPy 1.1x · Intermediate · ⏱ 120 min

By the end of this lesson you can

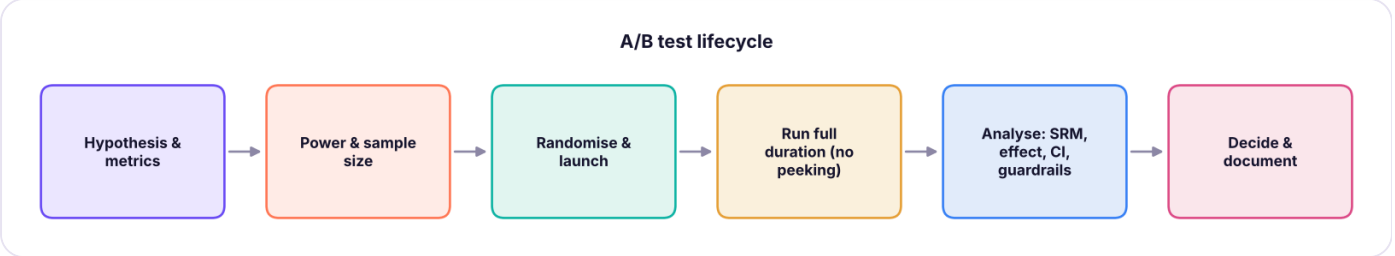
- Design an experiment with a hypothesis, primary metric, guardrail metrics and randomisation unit
- Compute the required sample size with power analysis and a minimum detectable effect
- Analyse conversion (two-proportion z-test) and revenue per user (Welch t-test and bootstrap)
- Check sample-ratio mismatch and recognise peeking, novelty and network effects
- Write a launch/no-launch recommendation with uncertainty and business impact

Datasets used

`ab_test.csv`

What is an A/B test?

A **randomised controlled experiment**: users are randomly split into **control (A)** and **treatment (B)**; the only systematic difference is the change you test, so differences in outcomes are **caused** by the change (up to chance, which statistics quantifies).



The experiment lifecycle.

> **Case.** An online store tests a **new one-page checkout (B)** against the current checkout (A). 4,000 users were split 50/50. **Hypothesis:** the new checkout increases the purchase conversion rate. **Primary metric:** conversion = purchasers ÷ users. **Secondary:** revenue per user, average order value. **Guardrails:** refund rate, page load time, support tickets.

Design choice	Good practice
Randomisation unit	User (not session or page view), so one person sees one version
Split	50/50 maximises power
Duration	At least one full business cycle (1–2 weeks, including weekends); fixed in advance
Metric	Defined before launch; one primary metric
Population	Who is eligible (new users only? all platforms?)

Sample size and power

Before launching: how many users do we need to detect the smallest effect worth acting on (the **minimum detectable effect**, MDE)?

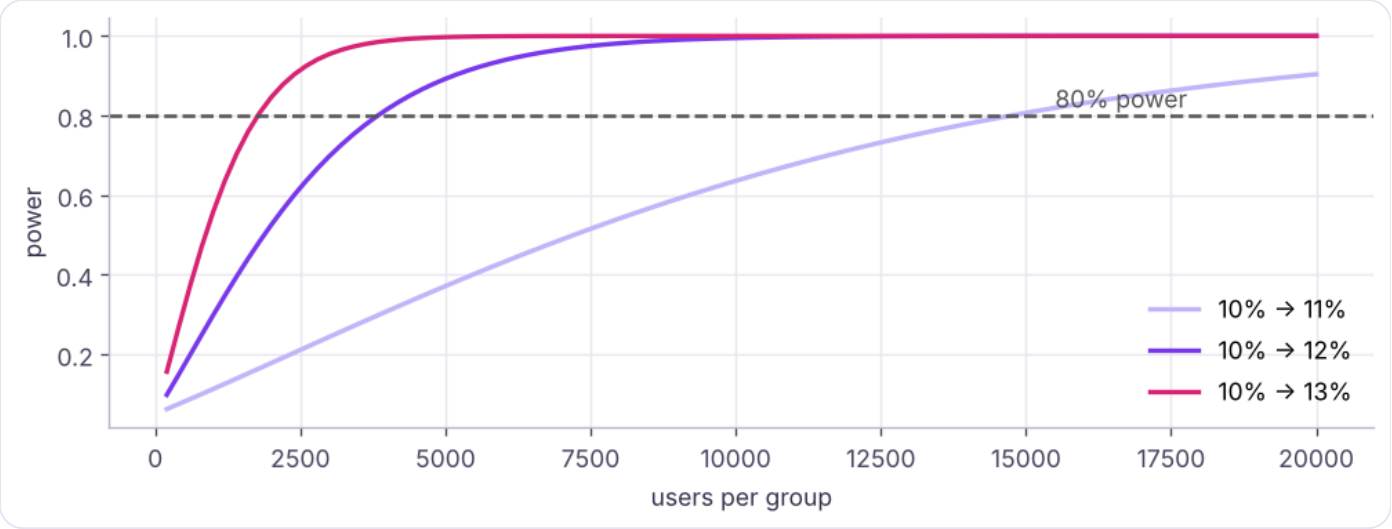
Python

```
import numpy as np, pandas as pd
from scipy import stats
from statsmodels.stats.power import NormalIndPower
from statsmodels.stats.proportion import proportion_effectsize
base = 0.10
for target in (0.11, 0.12, 0.13):
    h = proportion_effectsize(target, base)                    # Cohen's h
    n = NormalIndPower().solve_power(effect_size=h, alpha=0.05, power=0.8, ratio=1,
alternative="two-sided")
    print(f"detect {base:.0%} → {target:.0%} (+{(target - base) / base:.0%} relative):
{int(np.ceil(n)):,} users per group")
```

OUTPUT

```
detect 10% → 11% (+10% relative): 14,745 users per group
detect 10% → 12% (+20% relative): 3,835 users per group
detect 10% → 13% (+30% relative): 1,769 users per group
```

Small lifts need very large samples. With 2,000 users per group, this test can reliably detect a lift from about 10% to 13%, not smaller ones.



Power curves: the chance of detecting a true lift grows with sample size and effect size.

Analysis step 1: sanity checks

Python

```python
ab = pd.read_csv("ab_test.csv")
counts = ab.group.value_counts().sort_index(); print(counts.to_dict())
srm = stats.chisquare(counts.values)                                    # expected 50/50
print(f"sample-ratio mismatch test p = {srm.pvalue:.3f} → {'OK' if srm.pvalue > 0.01 else
'INVESTIGATE: randomisation or logging problem'}")
print("duplicate users:", ab.user_id.duplicated().sum(), "| missing:",
ab.isna().sum().sum())
```

OUTPUT

```
{'A_control': 2000, 'B_new_checkout': 2000}
sample-ratio mismatch test p = 1.000 → OK
duplicate users: 0 | missing: 0
```

A **sample-ratio mismatch (SRM)** (for example 52/48 when you planned 50/50, with $p < 0.01$) signals a bug in assignment or tracking; do not trust results until it is explained.

Analysis step 2: conversion rate

Python

```
from statsmodels.stats.proportion import proportions_ztest, confint_proportions_2indep
g = ab.groupby("group").agg(users=("user_id", "size"), buyers=("converted", "sum"))
g["conversion"] = g.buyers / g.users
print(g)
z, p = proportions_ztest(g.buyers.values[::-1], g.users.values[::-1])          # B vs A
lo, hi = confint_proportions_2indep(g.buyers.iloc[1], g.users.iloc[1], g.buyers.iloc[0],
g.users.iloc[0], compare="diff")
pa, pb = g.conversion.iloc[0], g.conversion.iloc[1]
print(f"absolute lift {100 * (pb - pa):.2f} pts (95% CI {100 * lo:.2f} to {100 * hi:.2f}) |
relative lift {(pb - pa) / pa:.1%} | z = {z:.2f}, p = {p:.5f}")
```

OUTPUT

```
                users  buyers  conversion
group
A_control        2000     198      0.0990
B_new_checkout   2000     273      0.1365
absolute lift 3.75 pts (95% CI 1.75 to 5.75) | relative lift 37.9% | z = 3.68, p =
0.00023
```

Analysis step 3: revenue per user and order value

Python

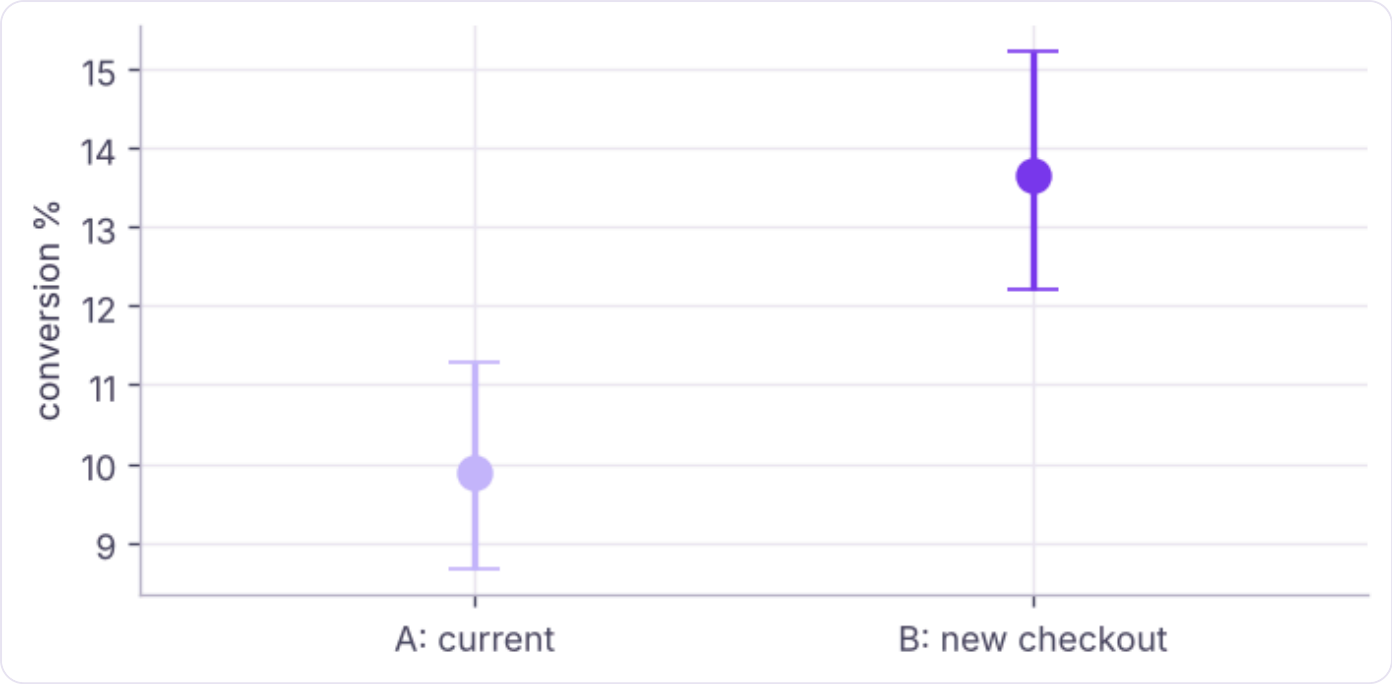
```
a, b = ab[ab.group == "A_control"], ab[ab.group == "B_new_checkout"]
for name, col, fa, fb in [("revenue per user", "order_value", a, b), ("average order value
(buyers)", "order_value", a[a.converted == 1], b[b.converted == 1])]:
    t = stats.ttest_ind(fb[col], fa[col], equal_var=False)
    print(f"{name:30s} A {fa[col].mean():8.1f} | B {fb[col].mean():8.1f} | diff
{fb[col].mean() - fa[col].mean():7.1f} | Welch p = {t.pvalue:.4f}")
rng = np.random.default_rng(0)
boot = [rng.choice(b.order_value.values, len(b)).mean() - rng.choice(a.order_value.values,
len(a)).mean() for _ in range(5000)]
print(f"bootstrap 95% CI for revenue-per-user difference: {np.percentile(boot, 2.5):.1f} to
{np.percentile(boot, 97.5):.1f}")
```

OUTPUT

```
revenue per user               A    137.5 | B    202.8 | diff    65.3 | Welch p =
0.0000
average order value (buyers)   A   1389.3 | B   1485.6 | diff    96.4 | Welch p =
```

```
0.0082
bootstrap 95% CI for revenue-per-user difference: 35.5 to 94.9
```

Revenue per user mixes conversion and order size and has many zeros and a long tail; Welch's t-test (thanks to the CLT) and the bootstrap both work at this sample size.



Conversion rate with 95% confidence intervals for each variant.

Analysis step 4: business impact

Python

```python
monthly_users = 300_000
extra_buyers = monthly_users * (pb - pa)
extra_rev = monthly_users * (b.order_value.mean() - a.order_value.mean())
print(f"at {monthly_users:,} users/month: about {extra_buyers:,.0f} extra orders and ₹{extra_rev / 1e5:,.1f} lakh extra revenue per month")
print(f"conservative (lower CI bound of conversion lift): {monthly_users * lo:,.0f} extra orders")
```

OUTPUT

```
at 300,000 users/month: about 11,250 extra orders and ₹195.8 lakh extra revenue per month
conservative (lower CI bound of conversion lift): 5,263 extra orders
```

Pitfalls

Pitfall	What goes wrong	Fix
Peeking	Checking daily and stopping at the first p < 0.05 inflates false positives to 20–30%	Fix the sample size; or use sequential tests / always-valid p-values
Too many metrics	Some will be "significant" by chance	One primary metric; adjust others (lesson 5)
Novelty / primacy effects	Users react to change itself, then revert	Run longer; analyse by week
SRM	Broken randomisation or logging	Chi-square check before analysis
Network effects	Treated users affect control users (marketplaces, social)	Cluster or geo randomisation
Simpson's paradox	Mix of segments differs between periods	Randomise; compare within segments
Underpowered tests	"No effect" just means "too small a test"	Power analysis first; report CIs
HARKing	Choosing the hypothesis after seeing results	Pre-register the plan

Python

```
rng = np.random.default_rng(11); false_wins = 0
for sim in range(400):                                    # A/A tests: there is NO
real difference
    xa, xb = rng.random(4000) < 0.10, rng.random(4000) < 0.10
    for day in range(1, 15):                              # peek every "day" (≈285
users per group per day)
        n = day * 285
        if proportions_ztest([xa[:n].sum(), xb[:n].sum()], [n, n])[1] < 0.05: false_wins +=
1; break
print(f"stopping at the first significant daily peek: {false_wins / 400:.0%} false
positives (should be 5%)")
```

OUTPUT

```
stopping at the first significant daily peek: 20% false positives (should be 5%)
```

Beyond the classic test

- **CUPED / regression adjustment:** use pre-experiment behaviour as a covariate to cut variance (and required sample size) by 20–50%.
- **Bayesian A/B testing:** reports "probability that B beats A" and expected loss; intuitive for stakeholders.
- **Multi-armed bandits:** shift traffic to the winning variant during the test (good for short campaigns, worse for clean inference).
- **Quasi-experiments** when you cannot randomise: difference-in-differences, synthetic control, regression discontinuity, propensity matching.
- Platforms: Optimizely, Statsig, GrowthBook, Eppo, LaunchDarkly, Firebase A/B Testing.

The decision memo

KEY TAKEAWAYS

Experiment: new one-page checkout, 4,000 users, 50/50, no sample-ratio mismatch. **Result:** conversion rose from 9.9% to 13.7% (+3.75 points, 95% CI about +1.7 to +5.8; $p < 0.001$). Revenue per user rose from about ₹138 to ₹203, and buyers' average order value also rose. **Guardrails:** check refunds, load time and support tickets before rollout. **Decision:** launch B to all users, with a 5% holdout for four weeks to confirm the effect lasts.

Summary

KEY TAKEAWAYS

- Randomisation makes A/B tests causal; define the hypothesis, one primary metric, guardrails and the duration before launch.
- Power analysis sets the sample size for a minimum detectable effect; small lifts need large samples.
- Check sample-ratio mismatch first; analyse conversion with a two-proportion z-test and revenue with Welch's t-test or the bootstrap; report CIs and business impact.
- Avoid peeking, metric fishing and underpowered tests; consider CUPED, Bayesian methods and quasi-experiments where appropriate.

Quick quiz

6 questions · tap an answer to check it

1. Why randomise users into A and B?

A So the only systematic difference between groups is the change being tested

B To make the test faster

C To increase conversion

D To avoid statistics

Answer: A. Randomisation balances all other factors on average.

2. What does a power analysis give you before the test?

A The p-value

B The sample size needed to detect a given minimum effect with a chosen probability

C The winner

D The conversion rate

Answer: B. Typical settings: $\alpha = 0.05$, power = 0.8.

3. The planned split was 50/50 but you observe 53/47 with SRM $p < 0.001$. What do you do?

A Report the result

B Investigate the assignment or logging bug before trusting the results

C Increase α

D Remove users until it is 50/50

Answer: B. SRM invalidates naive analysis.

4. Why is stopping as soon as p < 0.05 ("peeking") a problem?

- **A** It inflates the false-positive rate well above 5%
- **B** It wastes traffic
- **C** It makes p-values larger
- **D** It is fine

Answer: A. Use fixed horizons or sequential methods.

5. Which test suits comparing conversion rates of A and B?

- **A** Paired t-test
- **B** Two-proportion z-test (or chi-square)
- **C** ANOVA
- **D** Pearson correlation

Answer: B. Conversion is a proportion.

6. A test with 300 users per group shows no significant difference. Best conclusion?

- **A** The change has no effect
- **B** The test may be underpowered; look at the confidence interval for the plausible range of effects
- **C** Launch anyway
- **D** The metric is wrong

Answer: B. Absence of evidence is not evidence of absence.

EXERCISE

Practice

(1) Compute the users per group needed to detect a lift from 13.7% to 15% conversion. (2) Analyse average order value among buyers with a Mann-Whitney test as well. (3) Simulate an A/A test 1,000 times with a fixed horizon and confirm about 5% false positives. (4) Draft a one-paragraph memo for the case where the result had been p = 0.20.

Show solution (worked answer)

Python

```
print(NormalIndPower().solve_power(proportion_effectsize(0.15, 0.137), alpha=0.05,
power=0.8))
print(stats.mannwhitneyu(b[b.converted == 1].order_value, a[a.converted ==
1].order_value))
fp = np.mean([proportions_ztest([(rng.random(4000) < .1).sum(), (rng.random(4000) <
.1).sum()], [4000, 4000])[1] < .05 for _ in range(1000)]); print(fp)
# Memo: "No significant difference (p = 0.20); CI ranges from -x to +y points, so we
cannot rule out a meaningful lift. Extend the test to N users or test a bolder
change."
```

INTERVIEW CORNER

Questions you may be asked

Q: How would you design an A/B test for a new recommendation widget? State the hypothesis and primary metric (e.g. revenue per user), guardrails (latency, returns), randomise by user, size it with power analysis, run full weeks, check SRM, analyse with CIs, and decide with business impact.

Q: Your test is significant on day 3. Do you stop? No: stopping early inflates false positives and novelty effects can fade; finish the planned duration or use a sequential design.

Q: What if you cannot randomise? Use quasi-experimental methods: difference-in-differences with a comparison group, synthetic control, regression discontinuity or matching, and state the assumptions.

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